

GlobalGeoTree NRW: Observations, Sources, Districts, and Common Names

AUTHOR

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1 Purpose

This report focuses on **Nordrhein-Westfalen (NRW / North Rhine-Westphalia)**. It answers four practical inspection questions:

1. How many GlobalGeoTree observations are located in NRW?
2. How many recorded species are located in NRW?
3. Which data sources contribute NRW observations?
4. How are tree observations distributed across NRW districts?

The report also includes an **optional common-name enrichment step**. Scientific names remain the primary identifier, but the script can try to add English and German common names using GBIF through the optional `rgbif` package. Results are cached in a local CSV file so the report does not need to query GBIF every time.

Important note: common names are not globally standardized. The report treats German and English names as *GBIF vernacular names where available*, not as guaranteed official names.

2 Setup

3 Load observations

Input data summary

item	value
Input file	GlobalGeoTree.csv
Rows in input file	6,263,345
Rows after coordinate/species cleaning	6,263,345
Countries after cleaning	221

4 Preview of the dataset

First 10 cleaned rows

sample_id	country_code	level0	level1_family	level2_genus	level3_species	location	source	speci
1	MW	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes formosissimus	Malawi	iNaturalist Research	5,5

	Broadleaf			formosissimus		Research-grade Observations	
2 MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes formosissimus	Mozambique	iNaturalist Research-grade Observations	5,5
3 MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes sessiliflorus	Mozambique	iNaturalist Research-grade Observations	7,2
4 MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes sessiliflorus	Mozambique	iNaturalist Research-grade Observations	7,2
5 TZ	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia marina	Tanzania, United Republic of	iNaturalist Research-grade Observations	2,9
6 KE	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Kenya	naturgucker	2,9
7 NG	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Nigeria	Biodiversity of Some Coastal Communities in Akwa-Ibom State, Nigeria	2,9
8 NG	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Nigeria	Biodiversity of Some Coastal Communities in Akwa-Ibom State, Nigeria	2,9
9 NG	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Nigeria	Flora of Coastal Environment in Cross-River State, Nigeria	2,9
10 NG	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Nigeria	Flora of Coastal Environment in Cross-River State, Nigeria	2,9

5 Load NRW boundaries

6 Boundary data status

Boundary data status	
item	value
NUTS1 boundary status	Loaded NUTS1 from data/external/germany_nuts1_2021.geojson
NUTS3 boundary status	Downloaded NUTS3 from GISCO.
NRW boundary available	Yes
NRW district polygons	53

6 Filter observations to NRW

If NUTS boundaries are available, the report identifies NRW observations by spatially joining German observations to the NRW polygon. If boundaries are not available, it falls back to an approximate NRW bounding box.

NRW summary	
item	value
Filter method	NRW observations were identified using the official NUTS1 NRW boundary.
NRW observations	36,534
NRW recorded species	243
NRW recorded genera	126
NRW recorded families	61
Years represented	2015–2024

7 Optional: add Rare/Common/Frequent categories

The full GlobalGeoTree file may not contain **category** . If not, this report tries to add category labels from the larger evaluation file. This only labels species present in the evaluation lookup, so category coverage may be incomplete.

Category coverage in NRW	
item	value
Category method	Category was joined from GlobalGeoTree-10kEval-900.csv by species_key where possible, then by scientific species name.
NRW observations with known category	11,848
NRW species with known category	80

8 Optional: add English and German common names

This section first checks for a local cache file `species_common_names_gbif.csv` . If common names for some NRW species are missing and `rgbif` is installed, the report tries to retrieve English and German vernacular names from GBIF and updates the cache.

If this section is too slow, set `enable_gbif_lookup <- FALSE` in the setup chunk. The report will still render using scientific names.

Common-name coverage	
item	value
Common-name method	Common names were updated using GBIF and cached in species_common_names_gbif.csv
NRW species total	243
NRW species with English common name	242
NRW species with German common name	232

9 NRW observations and species

NRW observations and taxonomic coverage	
metric	value
Observations	36,534
Recorded species	243
Recorded genera	126
Recorded families	61
Data sources	16

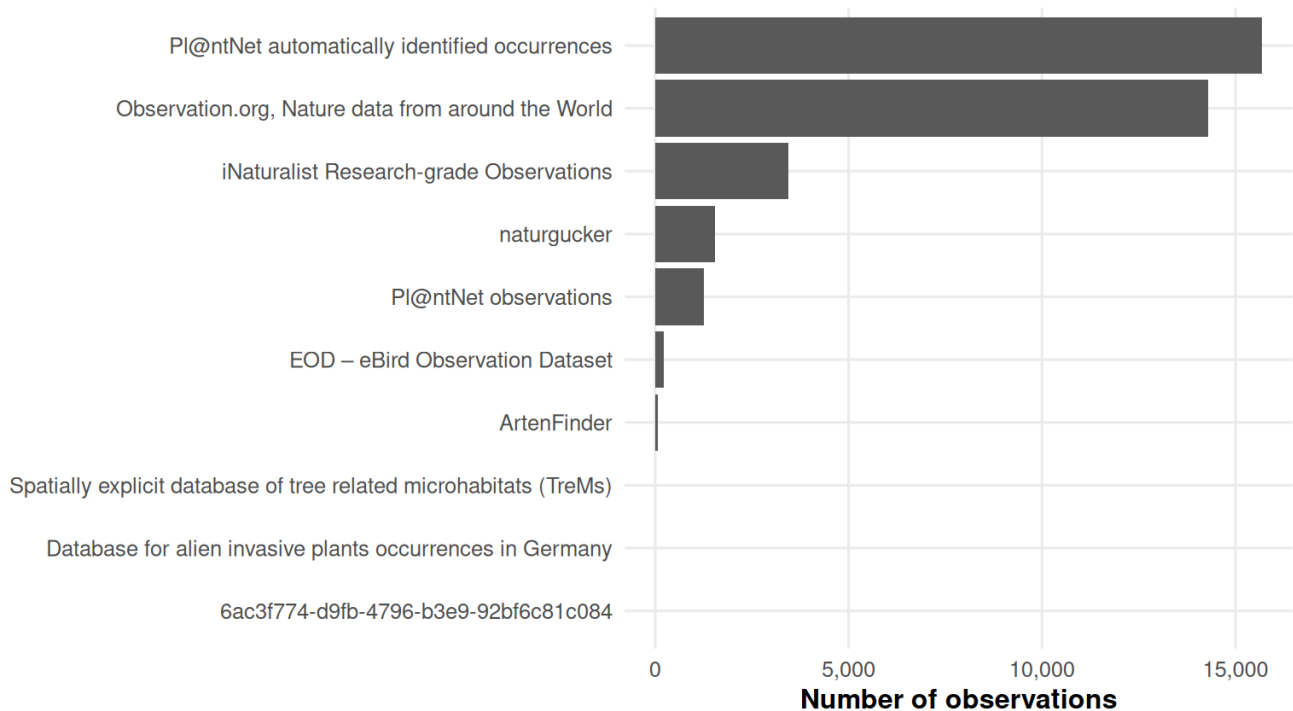
10 Sources of NRW observations

Top 15 data sources in NRW		
source	observations	share
Pl@ntNet automatically identified occurrences	15,693	43.0%
Observation.org, Nature data from around the World	14,288	39.1%
iNaturalist Research-grade Observations	3,436	9.4%
naturgucker	1,556	4.3%
Pl@ntNet observations	1,250	3.4%
EOD – eBird Observation Dataset	222	0.6%

ArtenFinder	66	0.2%
Spatially explicit database of tree related microhabitats (TreMs)	6	0.0%
6ac3f774-d9fb-4796-b3e9-92bf6c81c084	3	0.0%
Database for alien invasive plants occurrences in Germany	3	0.0%
14d5676a-2c54-4f94-9023-1e8dcd822aa0	2	0.0%
8a863029-f435-446a-821e-275f4f641165	2	0.0%
Anymals+plants - Citizen Science Data	2	0.0%
Birda - Global Observation Dataset	2	0.0%
Xeno-canto - Bird sounds from around the world	2	0.0%

Main sources of tree observations in NRW

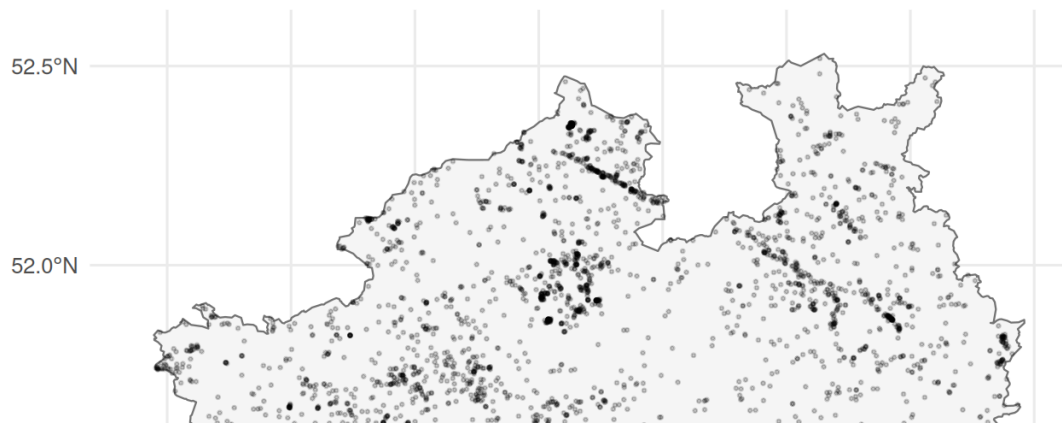
The plot shows the ten largest sources by observation c

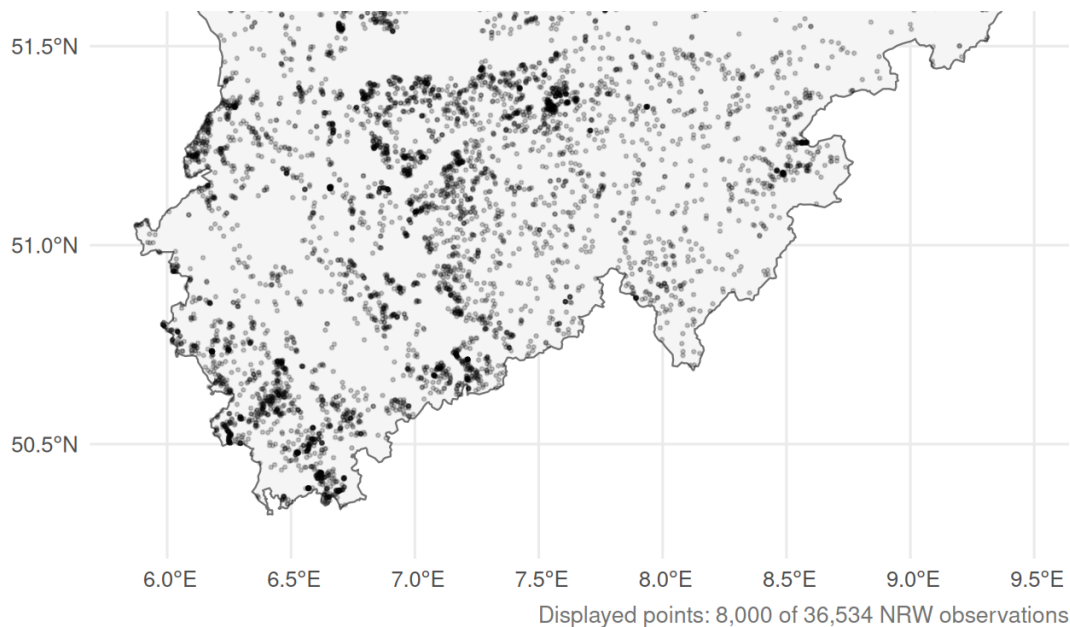


11 NRW observation map

Spatial distribution of GlobalGeoTree observations in NRW

Each point is one observation. Points are sampled for readability if the dataset is large.





12 NRW districts: observations and species

The district analysis uses **NUTS3 regions** in NRW. In Germany these correspond approximately to Kreise and kreisfreie Städte.

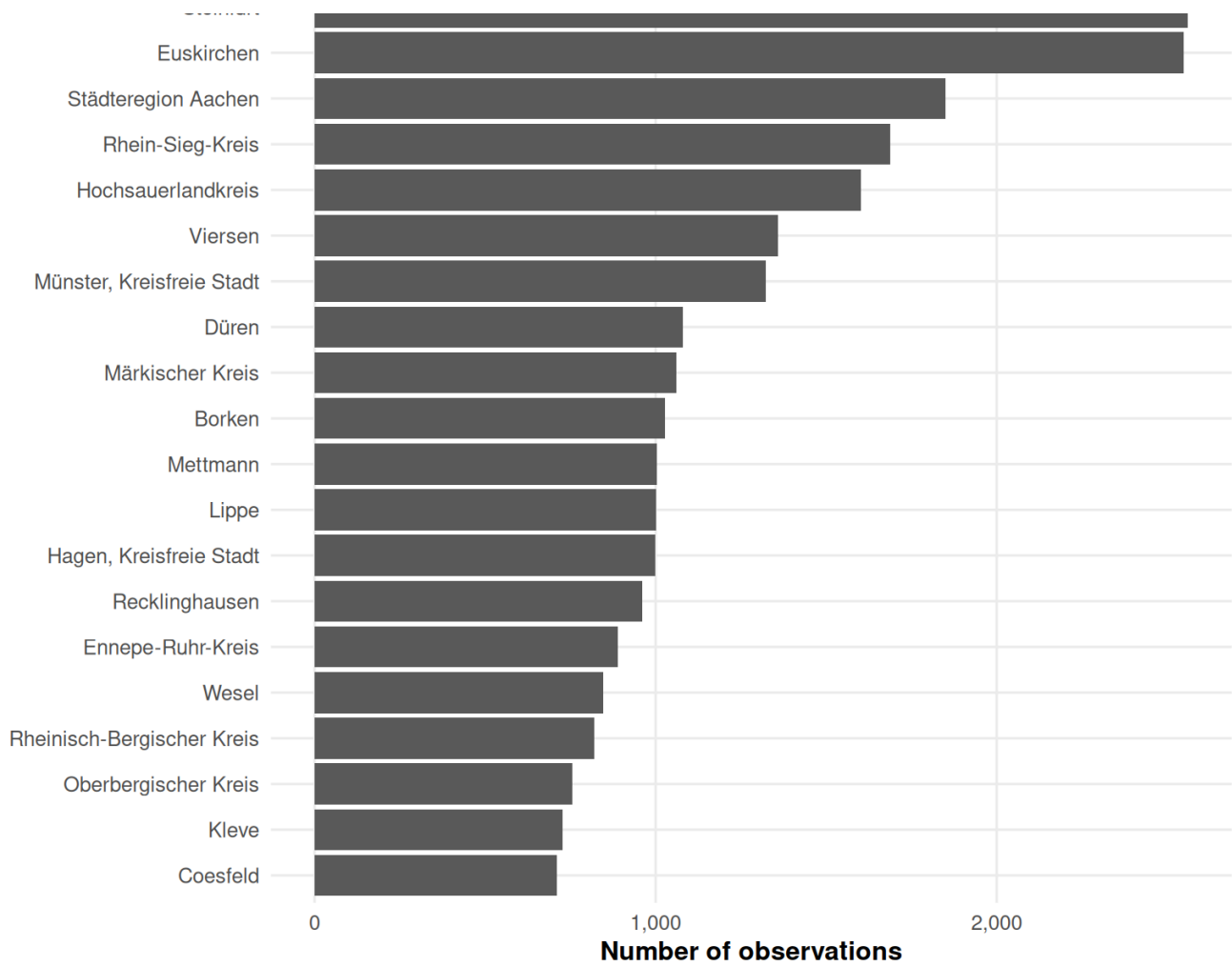
Top 15 NRW districts by observation count

district_id	district_name	observations	species	sources
DEA37	Steinfurt	2,561	126	6
DEA28	Euskirchen	2,548	103	7
DEA2D	Städteregion Aachen	1,850	110	8
DEA2C	Rhein-Sieg-Kreis	1,688	106	7
DEA57	Hochsauerlandkreis	1,603	92	7
DEA1E	Viersen	1,358	106	8
DEA33	Münster, Kreisfreie Stadt	1,323	99	8
DEA26	Düren	1,081	84	7
DEA58	Märkischer Kreis	1,061	80	7
DEA34	Borken	1,028	98	6
DEA1C	Mettmann	1,004	98	6
DEA45	Lippe	1,001	85	7
DEA53	Hagen, Kreisfreie Stadt	998	79	6
DEA36	Recklinghausen	961	89	7
DEA56	Ennepe-Ruhr-Kreis	890	87	6

NRW observations by district

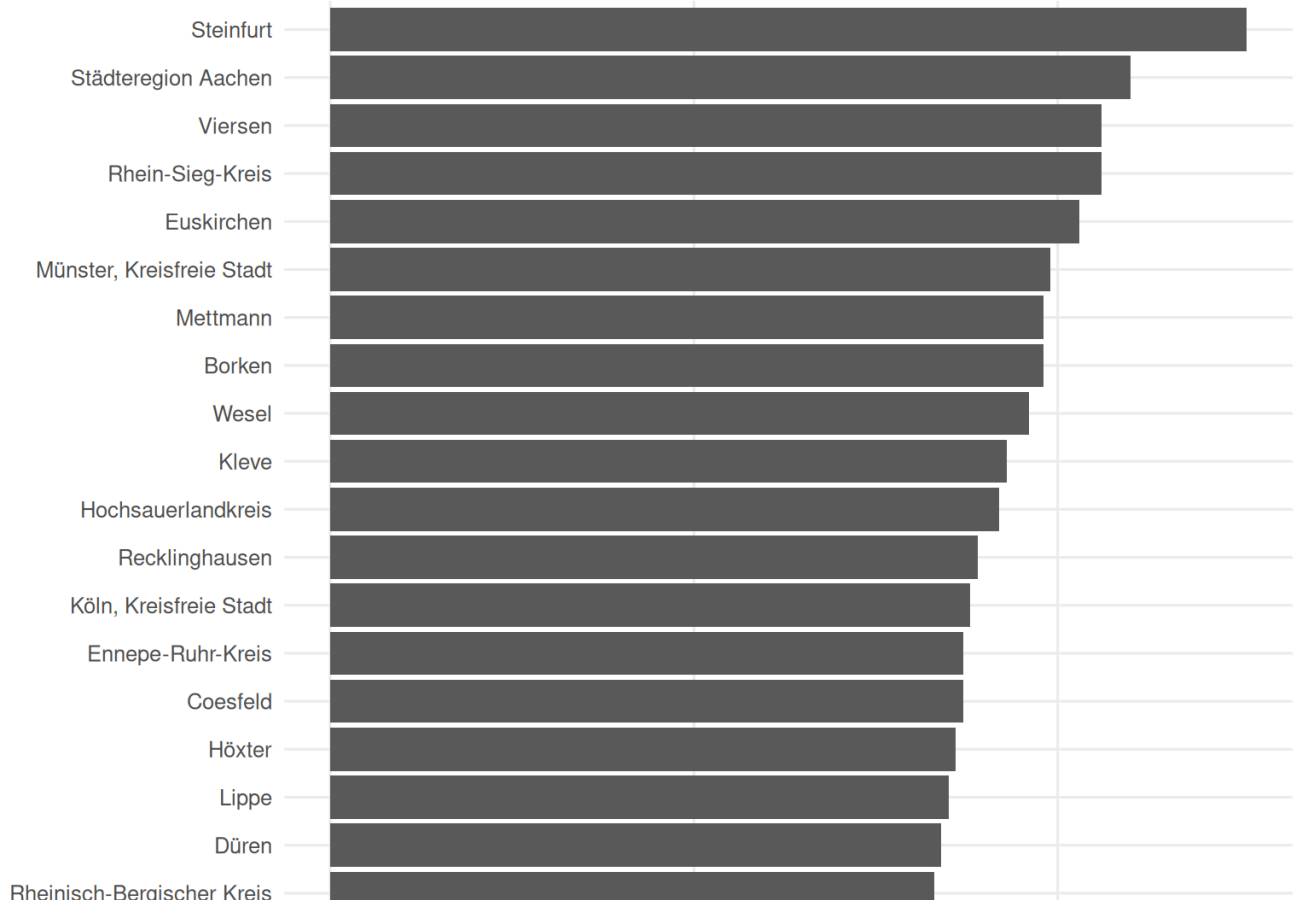
NUTS3 districts with the highest number of observations.

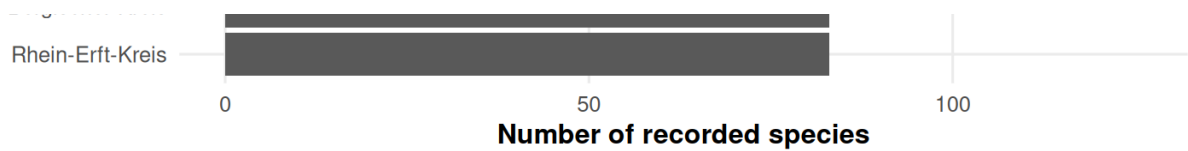
Steinfurt



Recorded species by NRW district

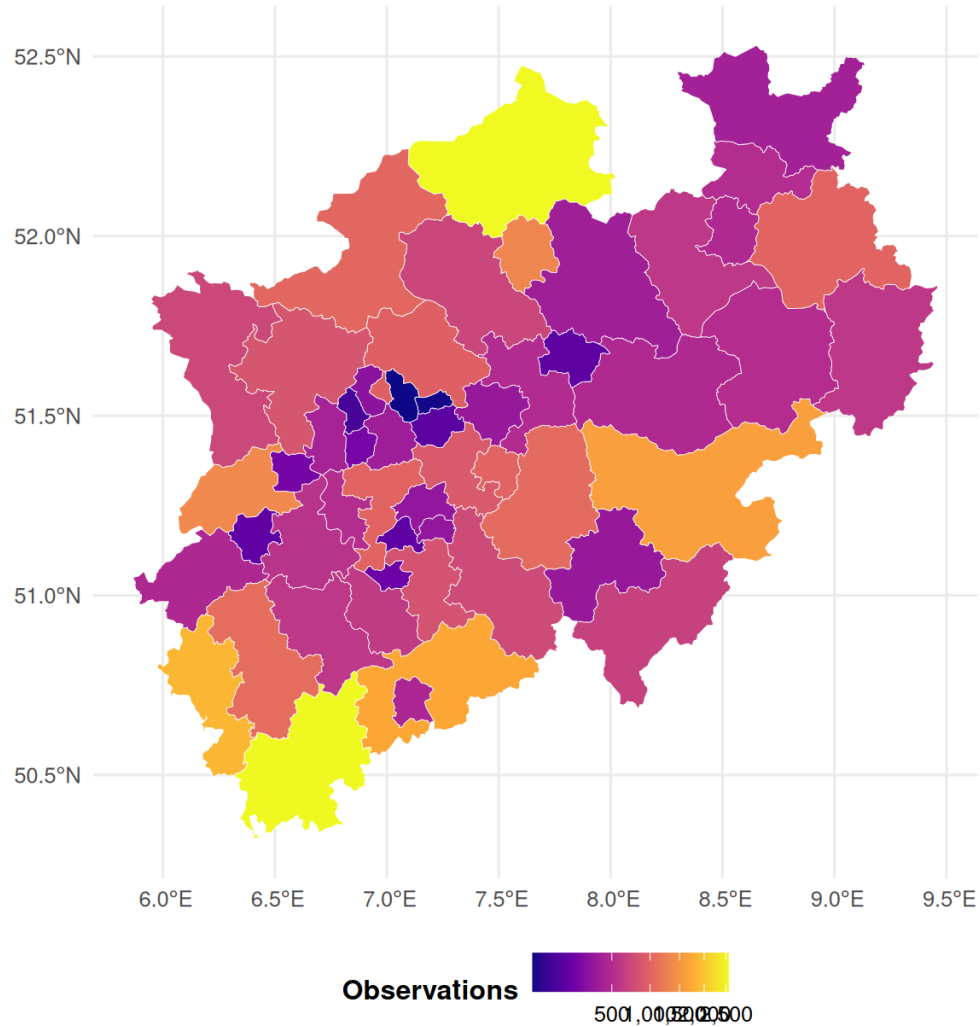
Districts with the highest number of distinct recorded species.





Observation frequency across NRW districts

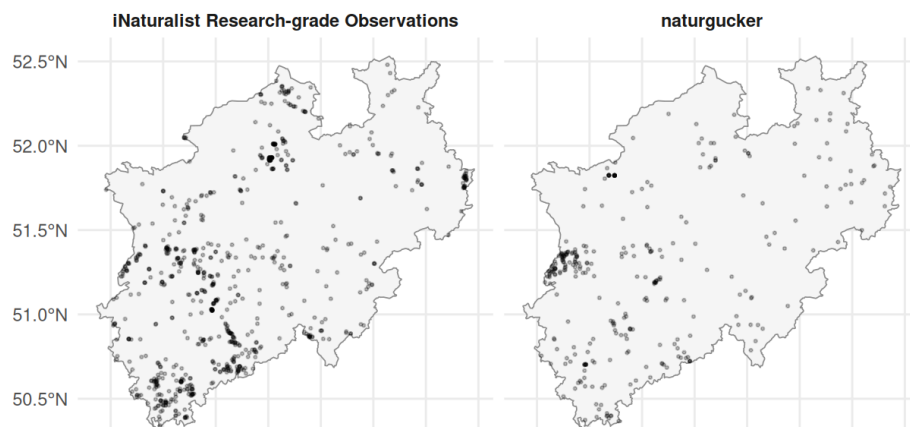
Darker districts contain more GlobalGeoTree observations. Square-root scaling improve

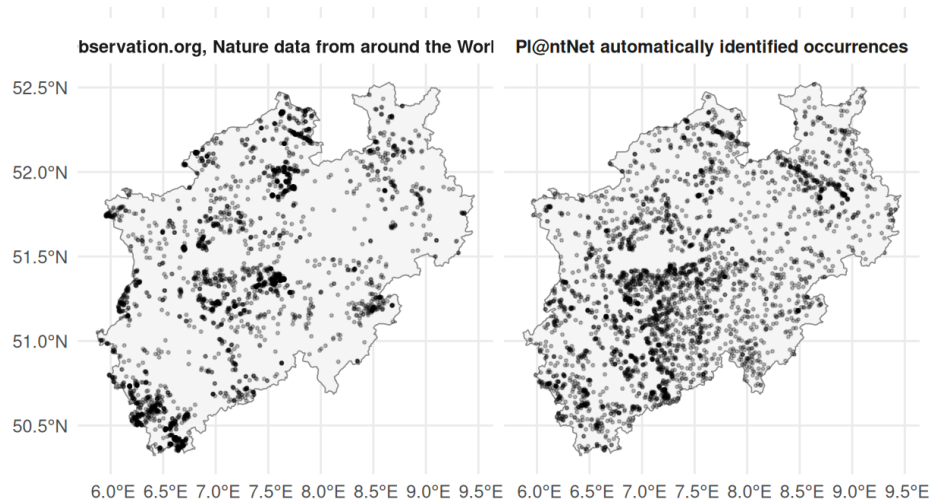


13 Sources on the NRW map

Spatial patterns of the largest observation sources in NRW

Facets show the four largest source groups. Points are sampled for readability.

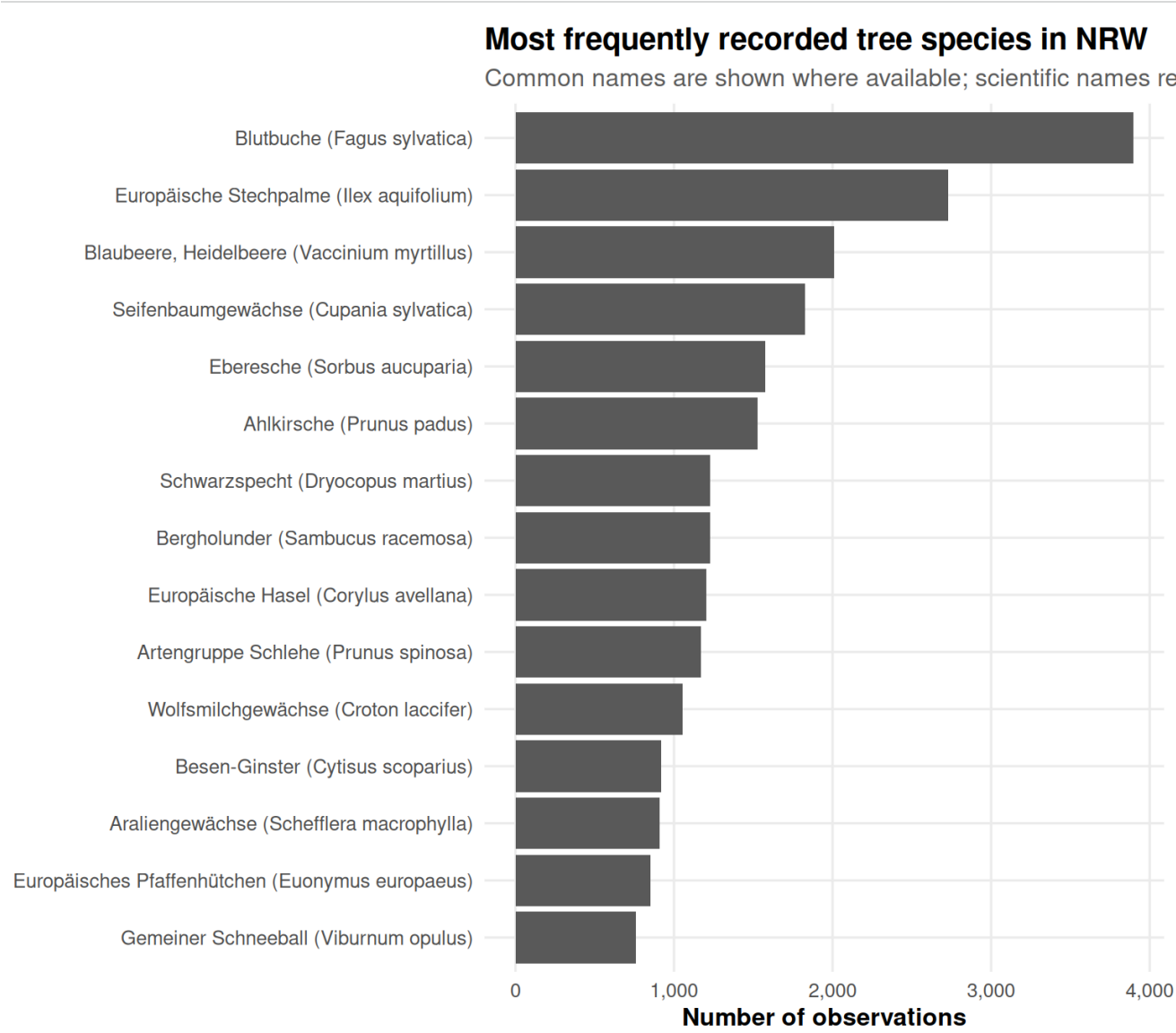




14 Top recorded species with common names

Top 20 recorded species in NRW					
display_name	scientific_name	german_common_name	english_common_name	observations	share
Blutbuche (Fagus sylvatica)	Fagus sylvatica	Blutbuche	Beech	3,896	10.7%
Europäische Stechpalme (Ilex aquifolium)	Ilex aquifolium	Europäische Stechpalme	English holly	2,728	7.5%
Blaubeere, Heidelbeere (Vaccinium myrtillus)	Vaccinium myrtillus	Blaubeere, Heidelbeere	Bilberry	2,011	5.5%
Seifenbaumgewächse (Cupania sylvatica)	Cupania sylvatica	Seifenbaumgewächse	Litchi family	1,827	5.0%
Eberesche (Sorbus aucuparia)	Sorbus aucuparia	Eberesche	European Mountain-ash	1,576	4.3%
Ahlkirsche (Prunus padus)	Prunus padus	Ahlkirsche	Bird Cherry	1,526	4.2%
Schwarzspecht (Dryocopus martius)	Dryocopus martius	Schwarzspecht	Black Woodpecker	1,229	3.4%
Bergholunder (Sambucus racemosa)	Sambucus racemosa	Bergholunder	American red elderberry	1,228	3.4%
Europäische Hasel (Corylus avellana)	Corylus avellana	Europäische Hasel	Barcelona-nuts	1,203	3.3%
Artengruppe Schlehe (Prunus spinosa)	Prunus spinosa	Artengruppe Schlehe	Blackthorn	1,172	3.2%
Wolfsmilchgewächse (Croton laccifer)	Croton laccifer	Wolfsmilchgewächse	Euphorbia family	1,053	2.9%
Besen-Ginster (Cytisus scoparius)	Cytisus scoparius	Besen-Ginster	Broom	917	2.5%
Araliengewächse (Schefflera macrophylla)	Schefflera macrophylla	Araliengewächse	Aralia and Ivy family	910	2.5%
Europäisches	Euonymus	Europäisches	Common Spindle	849	2.3%

Pfaffenhütchen (Euonymus europaeus)	europaeus	Pfaffenhütchen			
Gemeiner Schneeball (Viburnum opulus)	Viburnum opulus	Gemeiner Schneeball	Crampbark	761	2.1%
Eiche (Quercus oblongata)	Quercus oblongata	Eiche	Acorn	714	2.0%
Schwarzer Holunder (Sambucus nigra)	Sambucus nigra	Schwarzer Holunder	Black Elder	605	1.7%
Hagebuche (Carpinus betulus)	Carpinus betulus	Hagebuche	Caucasian Hornbeam	577	1.6%
Schwarz-Erle (Alnus glutinosa)	Alnus glutinosa	Schwarz-Erle	Alder	462	1.3%
Glockenheide (Erica excelsa)	Erica excelsa	Glockenheide	Ericas	440	1.2%



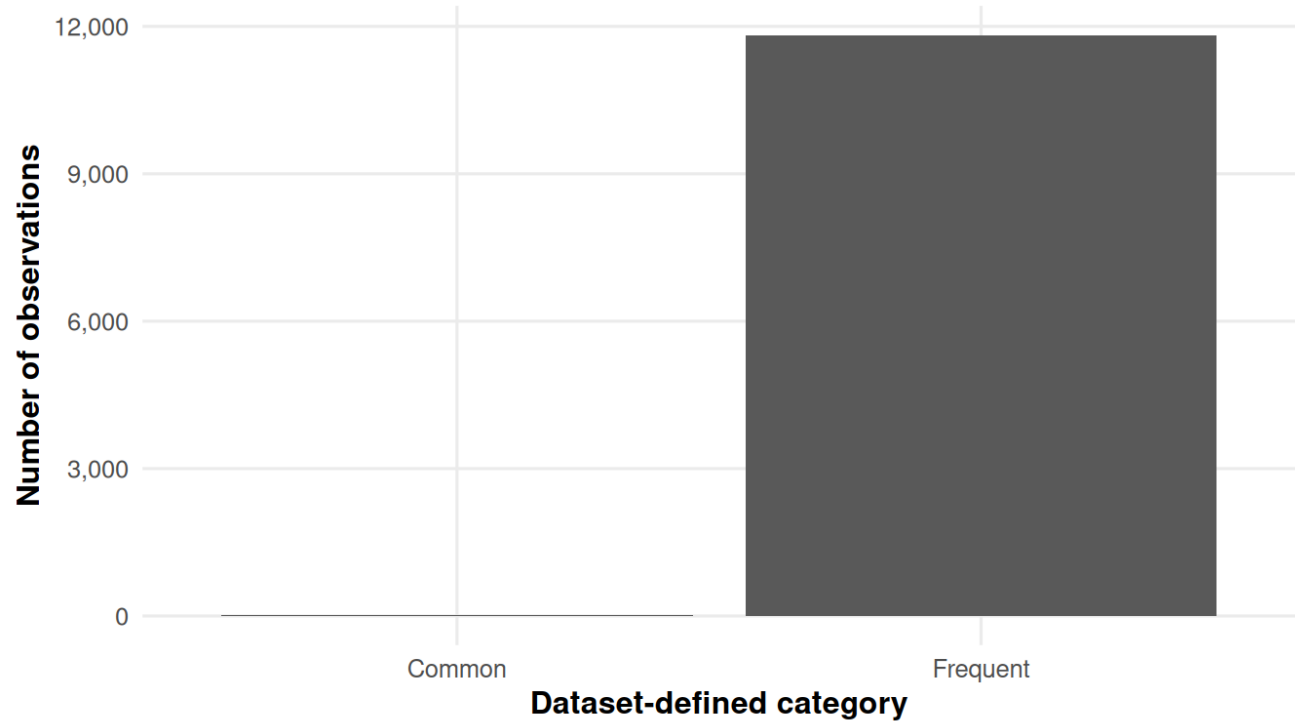
15 Category composition, if available

Rare/Common/Frequent coverage in NRW, where known

category	observations	species	observation_share
Frequent	11,825	77	99.8%
Common	23	3	0.2%

Known category composition of NRW observations

This only covers species that could be matched to a category lookup.



16 Interpretation notes

The NRW subset gives a regional view of GlobalGeoTree observations. The observation map and district-level charts describe where data are concentrated, but they should not be interpreted directly as true tree abundance. High counts may reflect a combination of tree presence, citizen-science activity, institutional data availability, and source-specific sampling behavior.

The common-name enrichment is useful for making the report more readable. However, common names are incomplete and not standardized across all languages or regions. For reproducibility, all analysis should continue to use scientific names or species keys as the stable identifiers.